

123/2515 王路路 Assignment 1

(1) For file A: $\lceil 1024 / 128 \rceil = 8$ blocks

For file B: $\lceil 500 / 128 \rceil = 4$ blocks

For file C: $\lceil 328 / 128 \rceil = 3$ blocks Total: 15 blocks

(2) Total storage: $15 \times 3 \times 128 \text{ MB} = 5760 \text{ MB}$

(3) $\lceil 300 / 128 \rceil = 2$ The minimal number of datanodes: $\lceil 45 / 2 \rceil = 23$

(4) Network traffic = Disk space = $(1000 \text{ MB} + 500 \text{ MB} + 328 \text{ MB}) \times 3 = 5556 \text{ MB}$

(5) It depends on the failed DataNode. In the worst case, network traffic = $2 \times 128 \text{ MB} = 256 \text{ MB}$